ALFIDO TECH INTERNSHIP

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Internship Domain: C / C++ Programming
Internship Duration: 10 April 2026 – 10 May 2026
Tools Used: GCC, G++, VS Code, GitHub, GDB

This internship project focused on strengthening programming fundamentals in C and C++ through practical implementation of data structures, algorithms, and system programming concepts. The tasks emphasized clean coding practices, logical problem solving, memory management, debugging, and real-world execution using professional development tools.

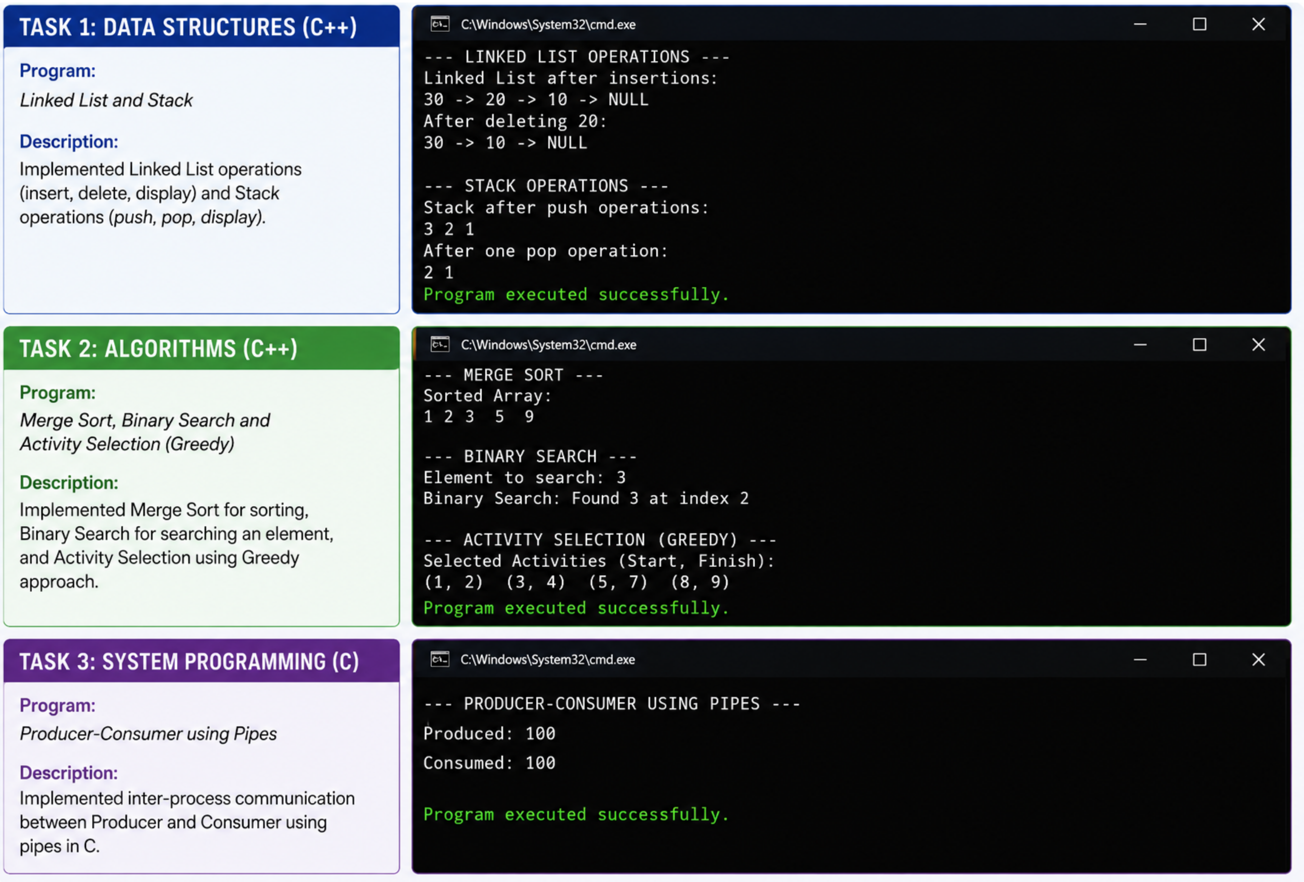
TASK 1: DATA STRUCTURES IMPLEMENTATION (C++)

Implemented core data structures including Linked List and Stack using C++. The project demonstrated insertion, deletion, traversal, push/pop operations, and memory handling. The objective of this task was to understand the internal working of dynamic data structures and improve logical implementation skills. **Key Features:**

- Linked List insertion and deletion operations
- Stack implementation using arrays
- Display functionality for visualization
- Complexity awareness and memory handling

Compile & Run Commands:

```
g++ main.cpp -o output
./output
```



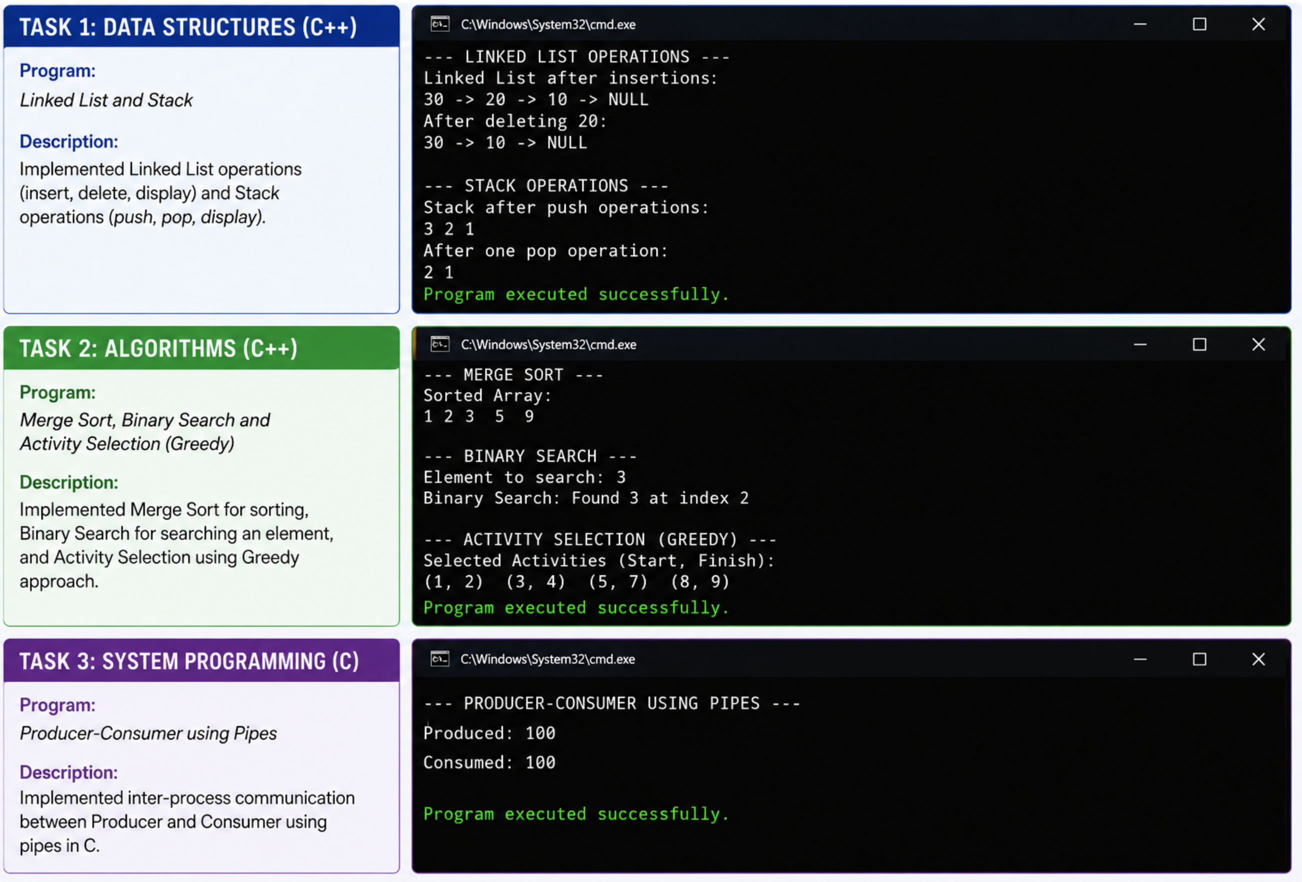
TASK 2: ALGORITHMS & PROBLEM SOLVING (C++)

Implemented important algorithmic techniques such as Merge Sort, Binary Search, and Greedy Activity Selection. This task focused on optimizing problem solving approaches and understanding time complexity. **Key Features:**

- Merge Sort using Divide & Conquer
- Binary Search for efficient searching
- Activity Selection using Greedy approach
- Runtime and complexity understanding

Compile & Run Commands:

g++ main.cpp -o output
./output



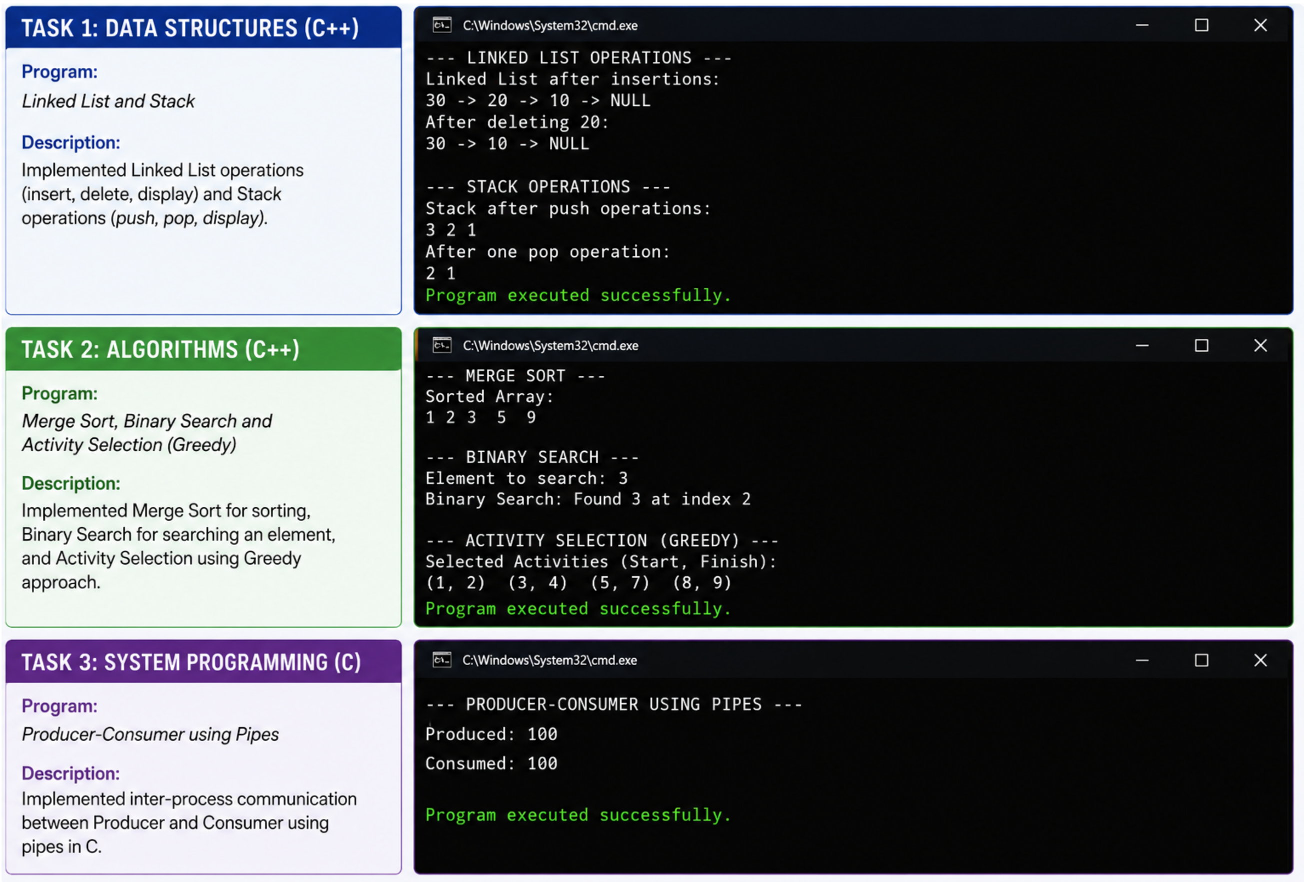
TASK 3: SYSTEM PROGRAMMING (C)

Developed a Producer-Consumer communication program using Pipes and Process Creation in C. This task introduced system-level programming concepts including inter-process communication and UNIX system calls. **Key Features:**

- Process creation using fork()
- Pipe-based communication
- Producer-Consumer implementation
- Error handling and process synchronization

Compile & Run Commands:

```
gcc main.c -o output
./output
```



CONCLUSION

The Alfido Tech Internship provided practical exposure to real-world C and C++ programming concepts. The tasks improved logical thinking, debugging capabilities, algorithmic understanding, and system-level programming knowledge. The project also enhanced software development practices such as documentation, GitHub management, and structured coding methodologies.